

MATHEMATICS

9709/11

Paper 1 Pure Mathematics 1 (P1)

May/June 2014

1 hour 45 minutes

Additional Materials: Answer Booklet/Paper
 Graph Paper
 List of Formulae (MF9)

READ THESE INSTRUCTIONS FIRST

If you have been given an Answer Booklet, follow the instructions on the front cover of the Booklet.

Write your Centre number, candidate number and name on all the work you hand in.

Write in dark blue or black pen.

You may use an HB pencil for any diagrams or graphs.

Do not use staples, paper clips, glue or correction fluid.

DO **NOT** WRITE IN ANY BARCODES.

Answer **all** the questions.

Give non-exact numerical answers correct to 3 significant figures, or 1 decimal place in the case of angles in degrees, unless a different level of accuracy is specified in the question.

The use of an electronic calculator is expected, where appropriate.

You are reminded of the need for clear presentation in your answers.

At the end of the examination, fasten all your work securely together.

The number of marks is given in brackets [] at the end of each question or part question.

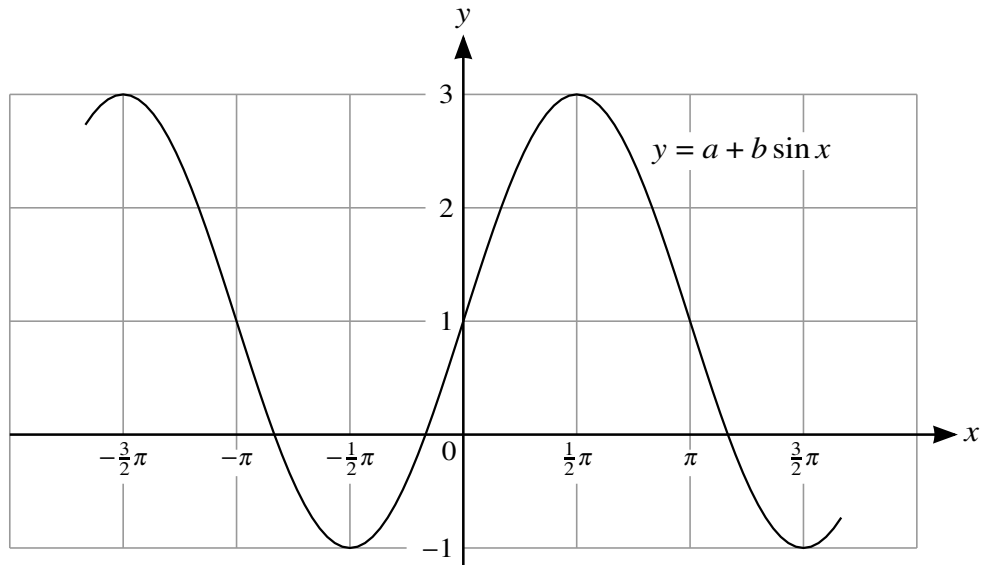
The total number of marks for this paper is 75.

Questions carrying smaller numbers of marks are printed earlier in the paper, and questions carrying larger numbers of marks later in the paper.

This document consists of 4 printed pages.



1



The diagram shows part of the graph of $y = a + b \sin x$. State the values of the constants a and b . [2]

2 (i) Express $4x^2 - 12x$ in the form $(2x + a)^2 + b$. [2]

(ii) Hence, or otherwise, find the set of values of x satisfying $4x^2 - 12x > 7$. [2]

3 Find the term independent of x in the expansion of $\left(4x^3 + \frac{1}{2x}\right)^8$. [4]

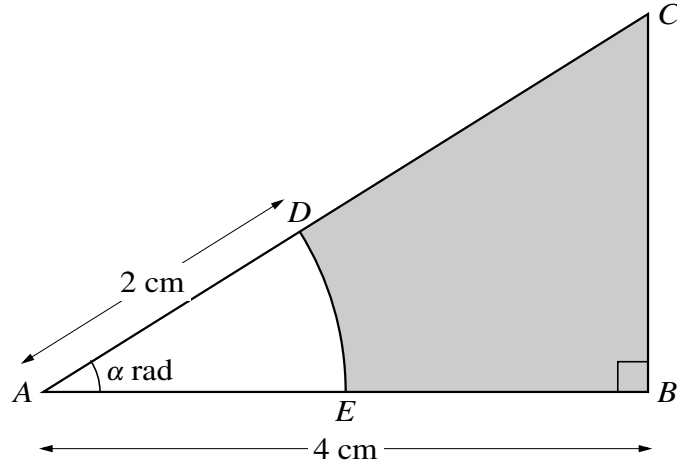
4 A curve has equation $y = \frac{4}{(3x + 1)^2}$. Find the equation of the tangent to the curve at the point where the line $x = -1$ intersects the curve. [5]

5 An arithmetic progression has first term a and common difference d . It is given that the sum of the first 200 terms is 4 times the sum of the first 100 terms.

(i) Find d in terms of a . [3]

(ii) Find the 100th term in terms of a . [2]

6



The diagram shows triangle ABC in which AB is perpendicular to BC . The length of AB is 4 cm and angle CAB is α radians. The arc DE with centre A and radius 2 cm meets AC at D and AB at E . Find, in terms of α ,

- (i) the area of the shaded region, [3]
- (ii) the perimeter of the shaded region. [3]

- 7 The coordinates of points A and B are $(a, 2)$ and $(3, b)$ respectively, where a and b are constants. The distance AB is $\sqrt{125}$ units and the gradient of the line AB is 2. Find the possible values of a and of b . [6]

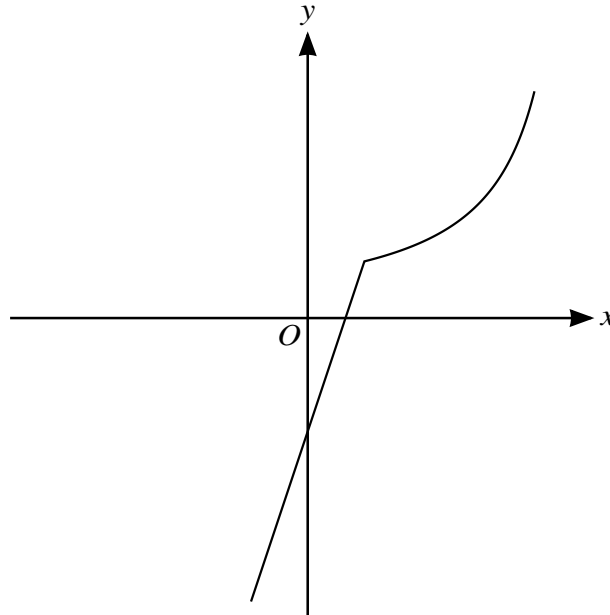
- 8 Relative to an origin O , the position vectors of points A and B are given by

$$\overrightarrow{OA} = \begin{pmatrix} 3p \\ 4 \\ p^2 \end{pmatrix} \quad \text{and} \quad \overrightarrow{OB} = \begin{pmatrix} -p \\ -1 \\ p^2 \end{pmatrix}.$$

- (i) Find the values of p for which angle AOB is 90° . [3]
- (ii) For the case where $p = 3$, find the unit vector in the direction of $\overrightarrow{BA}$. [3]

- 9 (i) Prove the identity $\frac{\sin \theta}{1 - \cos \theta} - \frac{1}{\sin \theta} \equiv \frac{1}{\tan \theta}$. [4]
- (ii) Hence solve the equation $\frac{\sin \theta}{1 - \cos \theta} - \frac{1}{\sin \theta} = 4 \tan \theta$ for $0^\circ < \theta < 180^\circ$. [3]

[Questions 10, 11 and 12 are printed on the next page.]



The diagram shows the function f defined for $-1 \leq x \leq 4$, where

$$f(x) = \begin{cases} 3x - 2 & \text{for } -1 \leq x \leq 1, \\ \frac{4}{5-x} & \text{for } 1 < x \leq 4. \end{cases}$$

- (i) State the range of f . [1]
- (ii) Copy the diagram and on your copy sketch the graph of $y = f^{-1}(x)$. [2]
- (iii) Obtain expressions to define the function f^{-1} , giving also the set of values for which each expression is valid. [6]

11 A line has equation $y = 2x + c$ and a curve has equation $y = 8 - 2x - x^2$.

- (i) For the case where the line is a tangent to the curve, find the value of the constant c . [3]
- (ii) For the case where $c = 11$, find the x -coordinates of the points of intersection of the line and the curve. Find also, by integration, the area of the region between the line and the curve. [7]

12 A curve is such that $\frac{dy}{dx} = x^{\frac{1}{2}} - x^{-\frac{1}{2}}$. The curve passes through the point $(4, \frac{2}{3})$.

- (i) Find the equation of the curve. [4]
- (ii) Find $\frac{d^2y}{dx^2}$. [2]
- (iii) Find the coordinates of the stationary point and determine its nature. [5]

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